

Dylan Butz

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Skills

Languages: Python, TypeScript, JavaScript, Java, C, SQL, HTML/CSS

Frameworks: React, FastAPI, LangChain, FAISS, Docling, NVIDIA NeMo Evaluator, vLLM, Triton Inference Server, pytest, Playwright

Infrastructure: Kubernetes, Docker, Ray, Airflow, Runpod, Cloudflare Workers, Keycloak, Prometheus, Grafana, GitHub Actions, Linux

Experience

JSI Telecom – Ottawa, ON

May 2026 - Present

AI Software Developer

- Built an automated LLM benchmarking platform integrating **NVIDIA NeMo Evaluator**, Airflow, and Ray, executing approximately **2,500 evaluation runs**
- Orchestrated inference across Hugging Face, NVIDIA Triton, and vLLM backends, using Ray's resource-aware scheduler to distribute tasks across CPU and GPU workers provisioned on demand through **Runpod**
- Secured a production vision analytics pipeline with Keycloak token-based authentication and role-based access control, automating infrastructure deployment through Kubernetes Jobs
- Developed **CI/CD** pipelines with GitHub Actions for production infrastructure, automating pytest-based testing, deployment, and validation workflows

NOKIA – Ottawa, ON

May 2025 – Aug 2025

Full Stack and AI/ML Developer

- Built a cloud-native **RAG agent** for 5G core error resolution, achieving **95% end-to-end accuracy** on evaluated error-code cases by retrieving supporting documentation and recommending the correct remediation
- Developed a multi-format ingestion pipeline combining Docling with document-specific parsers to index **3,000+ pages** for low-latency retrieval using FAISS vector search
- Built containerized MCP servers and **RESTful APIs** for automated cloud scaling using FastAPI, Docker, and Kubernetes

Projects

TogetherTune – YouTube Music Listening Parties

[github/TogetherTune](#)

- Launched a Chrome extension serving real users with synchronized YouTube Music playback, shared queues, invite links, and host-controlled permissions using **TypeScript**, React, and WXT
- Engineered an authoritative real-time backend with **Cloudflare Workers**, Durable Objects, KV, and WebSockets, supporting clock sync, automatic reconnection, session recovery, and host transfer
- Hardened the service with one-time connection tickets, rate limiting, runtime protocol validation, and CI-backed Vitest and **Playwright** coverage for two-client room flows

StudyBud

[github/StudyBud](#)

- Built a local-first Electron and React application that transforms course PDFs into structured topic divisions and generates page-cited answers, flashcards, and difficulty-based practice sets using OpenAI or Ollama
- Engineered a page-aware PDF.js pipeline that detects poor extraction, selectively invokes bundled PyMuPDF/Tesseract **OCR**, chunks recovered text, and preserves page provenance for citations
- Designed typed Electron IPC and SQLite/Drizzle persistence for documents and study artifacts, encrypted credentials with the OS keychain, and covered core workflows with **56 Vitest tests**

Education

Carleton University, **Software Engineering**

Sept 2024 – April 2028

- CGPA: 11 / 12
- Coursework: Object-Oriented Software Development, Algorithms and Data Structures
- Accolades: Deans Honour List, Entrance Scholarship, Golden Key Award (Top 15%)